

Nilesh Gupta

PhD Student
Department of Computer Science
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RESEARCH INTERESTS

My research focuses on End-to-end Information Retrieval and Efficient LLM architectures for very long-context modeling

EDUCATION

University of Texas at Austin 2021 - present
PhD student in Computer Science and Engineering; 4.00/4.00
Advisor: Prof. Inderjit Dhillon
Committee: Prof. Greg Durrett, Prof. Qiang Liu, Prof. Cho-jui Hsieh & Prof. Sewon Min

Indian Institute of Technology Bombay 2015 - 2019
B.Tech (Honours) in Computer Science and Engineering; 8.94/10.00
Advisor: Prof. Shivaram Kalyanakrishnan

WORK EXPERIENCE

Google Research Student Researcher, Fall 2024
Advisor: Dr. Felix Yu
Worked on developing BlockRank, a specialized long-context architecture for scalable LLM-based Ranking and RAG

Google Deepmind Student Researcher, Summer 2024
Advisor: Dr. Prateek Jain
Worked on developing *learned* super vocabulary tokenizers for large language models and establishing their scaling laws

Google Student Researcher, 2022-2023, Summer 2025
Advisor: Prof. Inderjit Dhillon
Worked on algorithms for jointly learning search index and representations, and recently on LLM-guided hierarchical retrieval

Microsoft Research India Research Fellow in Machine Learning and Optimization Group, 2019 - 2021
Advisor: Dr. Manik Varma
Worked on algorithms of Extreme Classification leading to multiple top-tier publications and impact across Microsoft products

PUBLICATIONS

Preprints

* - equal contribution

- **LUCID: Attention with Preconditioned Representations**
Sai-surya Duvvuri, Nirmal Patel, Nilesh Gupta, Inderjit Dhillon
Under review, 2025
- **LLM-guided Hierarchical Retrieval**
Nilesh Gupta, Wei-Cheng Chang, Ngot Bui, Cho-Jui Hsieh and Inderjit Dhillon
Under review, 2025
- **Compressing Many-Shots in In-Context Learning**
Devvrit Khatri, Pranamya Kulkarni, Nilesh Gupta, Yerram Varun, Liqian Peng, Jay Yagnik, Praneeth Netrapalli, Cho-Jui Hsieh, Alec Go, Inderjit S Dhillon, Aditya Kusupati and Prateek Jain
Under review, 2025

Conference Publications

- **Scalable In-context Ranking with Generative Models**
Nilesh Gupta, Chong You, Srinadh Bhojanapalli, Sanjiv Kumar, Inderjit Dhillon and Felix Yu
Neural Information Processing Systems (NeurIPS), 2025
- **EHI: End-to-end Learning of Hierarchical Index for Efficient Dense Retrieval**
Ramnath Kumar*, Anshul Mittal*, Nilesh Gupta, Aditya Kusupati, Inderjit Dhillon and Prateek Jain
Transaction of Machine Learning (TMLR), 2024
- **Exploring Design Choices for Building Language-Specific LLMs**
Atula Tejaswi*, Nilesh Gupta* and Eunsol Choi
Empirical Methods in Natural Language Processing EMNLP, 2024

- **Dual-encoders for Extreme Multi-label Classification**

Nilesh Gupta, Devvrit Khatri, Srinadh Bhojanapalli, Ankit S. Rawat, Prateek Jain and Inderjit Dhillon
International Conference on Learning Representations (ICLR), 2024

- **Negative Mining-aware Mini-batching for Extreme Classification**

Kunal Dahiya*, Nilesh Gupta*, Deepak Saini*, Akshay Soni, Yajun Wang, Kushal Dave, Jian Jiao, Gururaj, Prasenjit Dey, Amit Singh, Deepesh Hada, Vidit Jain, Bhawna Paliwal, Anshul Mittal, Sonu Mehta, Ramachandran Ramjee, Sumeet Agarwal, Purushottam Kar and Manik Varma
International Conference on Web Search and Data Mining (WSDM), 2023

- **ELIAS: End-to-end Learning to Index and Search in Large Output Spaces**

Nilesh Gupta, Patrick Chen, Hsiang-Fu Yu, Cho-jui Hsieh and Inderjit Dhillon
Neural Information Processing Systems (NeurIPS), 2022

- **Generalized Zero-Shot Extreme Multi-Label Learning**

Nilesh Gupta, Sakina Bohra, Yashoteja Prabhu, Saurabh Purohit and Manik Varma
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2021

- **Extreme Regression for Dynamic Search Advertising**

Yashoteja Prabhu, Aditya Kusupati, Nilesh Gupta and Manik Varma
International Conference on Web Search and Data Mining (WSDM), 2020 (**Long Oral**)
Workshop on eXtreme Classification: Theory and Applications @ ICML, 2020

SELECTED AWARDS AND HONORS

- Awarded \$50,000 Gemini Academic Program Award for research on LLM-guided retrieval
- Ranked 4th in ACM-ICPC Asia Regionals and 6th in ACM-ICPC India Online
- All India Rank 384 in JEE Advanced (IIT-JEE) 2015 among 150,000 candidates
- Awarded the prestigious KVPY Fellowship from Government of India
- Ranked 2nd in Regional Mathematics Olympiad (RMO) and among top 300 students in INMO

TEACHING & RESPONSIBILITIES

- *Graduate Teaching Assistantship* - Computer Science Department, UT Austin
 - Symbolic Programming - Prof. Gordon S. Novak Fall 2022
 - Fundamentals of Machine Learning - Prof. Inderjit Dhillon Spring 2023, Spring 2025
 - Principles of Machine Learning I - Prof. Angela Beasley Fall 2023, Spring 2024
- *Undergraduate Teaching Assistantship* - Computer Science and Engineering, IIT Bombay
 - Computer Programming and Utilisation - Prof. Ganesh Ramakrishnan Autumn 2018
 - Computer Programming and Utilisation - Prof. Krishna S. Autumn 2017
 - Basic Calculus - Prof. Amiya K. Pani Autumn 2016
- *MOOC Teaching Assistantship* - IITBombayX, edX
 - Data Structures and Algorithms - Prof. Deepak B. Phatak Spring & Autumn 2017
- *Managing Extreme Classification Reading Group* - Microsoft Research India 2020 - 2021